PROJECT MILESTONE 01

Proposal and Agent Specification

SmartCart AI

Grocery Shopping and Budget Optimization Agent

Selects groceries within a budget while considering quantity, household requirements, and prices.

Project Proposal

Course	Artificial Intelligence
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Milestone	Project Milestone 01: Proposal and Agent Specification
Mapped CLOs	CLO 1 and CLO 2

1. Refined AI Project Topic

SmartCart AI is an intelligent grocery shopping and budget optimization agent. Given a fixed budget, a household profile, and a list of required items, the agent selects which groceries to buy, in what quantities, and at what prices, so that household needs are met without exceeding the budget.

2. Problem Statement

Households often struggle to plan grocery purchases within a limited budget. Prices vary across products, household requirements differ by family size, and manual shopping leads to overspending, excess purchases, and missing essentials. An intelligent system is needed to select the right groceries in the right quantities while staying within the budget.

3. Objectives

- Select groceries such that the total cost never exceeds the budget.
- Decide appropriate quantities based on household size and requirements.
- Prioritize essential items over optional items when the budget is limited.
- Choose cost-effective options by comparing prices.
- Maximize the value of the basket while minimizing wasted spending.

4. Initial Project Concept

The agent takes the budget, household requirements, and an item price list as input. It treats grocery selection as a constrained optimization and search problem, adding, removing, or replacing items and adjusting quantities, and outputs an optimized shopping list with total cost and remaining budget.

5. PEAS Specification

Component	Description for SmartCart AI
Performance Measure	- Total cost stays within the budget - Percentage of household requirements and essentials satisfied - Overall value of the selected basket - Minimal wasted spending
Environment	- Grocery store or online catalogue with items and prices - User's budget - Household profile and requirements - Essential and optional items
Actuators	- Add item to cart - Remove or replace item - Increase or decrease quantity - Display final shopping list, total cost, and remaining budget
Sensors	- User input (budget, household size, requirements) - Item prices and product data - Current cart contents and total cost - Remaining budget

6. Task Environment Analysis

Property	Classification	Justification
Observability	Fully observable	The agent knows all prices, the budget, and the cart state.
Agents	Single agent	Only SmartCart AI makes the decisions.
Determinism	Deterministic	Adding an item changes the cart and budget predictably.
Episodic / Sequential	Sequential	Each selection affects the remaining budget and later choices.
Static / Dynamic	Static	Prices and requirements do not change during planning.
Discrete / Continuous	Discrete	Items, quantities, and actions are finite discrete choices.
Known / Unknown	Known	The rules (budget limit, price calculation) are known.

7. Agent Requirements

- **Agent type:** goal-based and utility-based agent.
- **Knowledge base:** grocery items with category, price, unit, and priority.
- **State:** current cart, quantities, total spent, and remaining budget.
- **Goal test:** essential needs met and total cost within budget.
- **Utility function:** value per unit cost, with higher weight for essentials.
- **Search / optimization:** greedy best-first or knapsack-style optimization.
- **Interface:** inputs for budget and requirements; output of final list and cost.